

CSMFAB000000000112

Aegis-Forge I: Assembly Line Integration, Output Slug Catalog, and Material Hand-off

Downstream Processing, Output Classification, and Seed Material Generation for CSMFAB Portfolio

Document Number: CSMFAB000000000112 (Volume 5 of 5) **Series:** Aegis-Forge Thermal Cascade Inductive Material Separator (TCIMS) **Date:** June 2026

1. The Output Slug Catalog: Every Exit Stream has a Destination

The Aegis-Forge I produces 12 distinct output streams per full thermal cascade cycle. Each is classified, sampled, stamped, and routed to its specific downstream use. None are waste. All are value.

1.1 Output Stream Registry

Stream #	Port	Name	Phase at Exit	Temperature	Downstream Use
S-01	1	Process Water	Liquid	80-100°C	AMLS recirculation
S-02	2	Mixed Polymer Slug	Solid (cooled)	25°C	Polymer reprocessing / Elium® feedstock
S-03	3	Carbon Char Pellets	Solid granule	25°C	Activated carbon / carburizing agent
S-04	4a	Indium-Rich Fraction	Liquid → Solid	157-200°C	Ionic liquid In refining → YInMn Blue
S-05	4b	Zn/Pb/Sn Slug	Solid ingot	25°C	Low-temp metal recovery / solder remanufacture
S-06	5	Aluminum Slug	Solid ingot	25°C	Al casting / secondary market sale
S-07	6	Copper Billet	Solid ingot	25°C	Cu wire remanufacture / electrical market
S-08	6-PM	Precious Metal Fraction	Included in S-07	—	Electrolytic Au/Ag/Pd refining
S-09	7	Ferrous Slug	Solid ingot	25°C	Fe casting / RE magnet leach (from NdFeB)

Stream #	Port	Name	Phase at Exit	Temperature	Downstream Use
S-10	7-RE	Rare Earth Concentrate	Included in S-09	—	Hydromet RE extraction □ magnets, catalysts Hydromet refinery □ Zr, Ti, REE recovery Combustion for Stage 7-8 auxiliary heat Drum purge gas / geopolymer curing atmosphere Construction material additive
S-11	8	Mineral Concentrate	Mixed solid	25°C	
S-12	Gas	Pyrolysis Fuel Gas	Gas	25°C	
S-13	Gas	CO2 Stream	Gas	25°C	
S-14	Gas	NaCl / CaSO4 Solids	Solid	25°C	

2. Slug Cooling Station

All liquid-metal slugs exit their respective ports into a carousel of ZrO₂ ceramic molds:

Carousel slug mold system: - 16-position rotary carousel (BFRP/Elium® disc, PEEK CF40 bushing) - Mold material: 3Y-TZP ZrO₂ ceramic — withstands all metal temperatures; negligible reaction - Mold release: none needed — ZrO₂ does not wet with molten metals (contact angle >120°) - Cooling: N₂ gas quench nozzles (PEEK CF40 manifold, CSMFAB000000000014-type ducting) Controlled cooling rate: 20-100°C/min (fast for fine grain Al; slow for ceramic-melts to reduce stress) - Mold capacity: 2 kg per slug mold (standard size is 100×50×25mm brick) - Cycle time per position: 8 minutes (fill + cool enough for demolding) - Automated demolding: PEEK CF40 ejector pins, KNbO₃-BaTiO₃ piezo actuated

Slug stamp-and-barcode system: Before cooling sets, a ZrO₂ ceramic stamp head presses an alphanumeric code into each slug: - Batch number, port of origin, stage temperature, date/time - This traceability is essential for Phoenix Protocol — knowing the full provenance of recycled material

3. Slag Analysis and Auto-Sorting

Each slug emerging from the carousel is analyzed by an inline spectrometer:

Laser-Induced Breakdown Spectroscopy (LIBS): - Nd:YAG pulsed laser (1064 nm, 5 mJ pulses) focused on slug surface - KNbO₃-BaTiO₃ detector vibrates synchronously (lock-in amplification of plasma emission) - PMMA POF fiber carries emission to spectrometer - Analysis time: 2 seconds per slug - Elements detectable: all metals, C, Si, S, Cu, Fe, Al, Zn, Pb, Ni, REE signatures - Result: automatic grade assignment and routing to appropriate downstream conveyor

Routing conveyors: Three output classifications per slug: 1. **Direct use:** Meets specification for known downstream CSMFAB process □ immediately conveyed to that machine 2. **Refinement needed:** Good material but needs purity improvement □ routed to refinery queue 3. **Secondary sale:** Not directly usable by CSM □ weighed, logged, routed to commercial sale

4. The CSM Seed Material Generation Map

This is the Aegis-Forge I's ultimate purpose: **converting raw Earth and recycled matter directly into the seed materials required for every CSMFAB product.**

4.1 Seed Material □ CSMFAB Product Direct Linkage

Aegis-Forge Output	Intermediate Step	Final CSMFAB Seed Material	Used In
S-04 In-rich fraction	Ionic liquid refining (Phoenix Protocol)	Indium metal 99.9%	YInMn Blue (35 products)
S-10 RE Concentrate	Hydromet leach □ Cyanex 272 extraction	Y ₂ O ₃ , Nd ₂ O ₃ , La ₂ O ₃	YSZ, YInMn, RE magnets
S-11 Mineral Concentrate	Leach + ZrO ₂ precip + Y ₂ O ₃ addition	3Y-TZP ZrO ₂ powder	ZrO ₂ fasteners, parts
S-11 (ZrSiO ₄ dissociation)	Carbothermal + SHS program	ZrB ₂ powder	CSMFAB0000000000001
S-11 (TiO ₂ concentrate)	Chloride-thermal extraction	TiO ₂ □ TiCl ₄ □ Ti ₃ AlC ₂	MXene, MAX Phase
S-03 Carbon Char	Activation (steam, 900°C)	Activated carbon	Air/gas filtration
S-03 Carbon Char	Graphitization (2800°C)	Graphite	Electrode material
S-07 Copper Billet	Melt + draw	Cu wire	Motor windings (isolated)
S-09 Ferrous Slug	RE extraction first, then melt	Clean iron + RE separated	Iron castings (tooling)
S-06 Aluminum Slug	Melt + refine	Casting alloy or scrap sale	Commercial (CSM uses no Al in Aegis products)
S-12 Pyrolysis gas	Steam reform (Option B)	H ₂ fuel	Engine
S-08 Precious metals	Electrolytic refining	Au, Ag, Pd	CSMFAB0000000000105
S-05 Zn/Pb/Sn	Fractional distillation	Separated Zn/Pb/Sn	Sale □ Funds operations
S-02 Polymer slug	Extrusion □ filament or sheet	Polymer blend stock	Sale or solder remanufacture
			Injection-moldable polymer

4.2 Critical Path: Rock □ ZrB₂ Powder in 48 Hours

This is the most important operational sequence showing the Aegis-Forge I as a true cradle-to-material system for the portfolio's most critical ceramic:

Hour 0: Zircon sand (ZrSiO₄) loaded into Aegis-Forge I AMLS
[20 min acoustic: >80% zircon grain liberation from silica matrix]

Hour 0.3: AMLS magnetic separation removes ilmenite fraction (Fe, Ti separate stream)

Hour 0.5: Liberated zircon sand fed to Forge drum
Stage 8 temperature: 1700°C
ZrSiO₄ dissociation: ZrSiO₄ □ ZrO₂ + SiO₂ (liquid SiO₂ exits Port 8 first)

Hour 3: Stage 8 mineral concentrate (ZrO₂-rich) exits as Slug S-11

Hour 3.5: S-11 transferred to Phoenix Protocol hydromet circuit:
 BFRP reactor vessel; H₂SO₄ leach → ZrO₂ dissolves as Zr⁴⁺
 YCl₃ added (Yttrium from REE stream S-10)
 NH₄OH co-precipitation → Zr-Y hydroxide gel

Hour 6: Gel filtered, washed, dried at 120°C

Hour 8: Calcination furnace (1200°C, 4h) → 3Y-TZP ZrO₂ crystalline powder

Hour 12: 3Y-TZP ZrO₂ powder available for:
 → Direct: use as ZrO₂ fastener/part powder (sintered at 1400°C)
 → Alternative: add to B₂O₃ + carbon → ZrB₂ via carbothermal synthesis

Hour 36: ZrB₂ carbothermal reaction complete (1700°C, 16h)
 OR: ZrB₂ SHS reaction (1h from Zr metal + B precursors)

Hour 48: ZrB₂ powder available for Flash Sintering into Aegis-C panels, stove burners, etc.

48 hours from raw zircon sand to ready-to-sinter ZrB₂ powder. No metals, no exotic chemistry, no supply chain vulnerabilities.

5. Downstream Assembly Line Integration

5.1 Aegis-Forge I as Machine No. 1

The Aegis-Forge I is explicitly the **first machine** in a modular assembly line. Output slugs flow via gravity-conveyor (BFRP belt conveyor on ZrO₂ ceramic rollers) to adjacent downstream machines. The complete line, when built, comprises:

Machine	CSMFAB Number	Function	Input from Forge
Aegis-Forge I	108-112	Thermal cascade separation	Raw feedstock
Hydro-Refinery I	113 (future)	Hydrometallurgical critical mineral extraction	S-11 mineral concentrate
Powder Synthesis I	114 (future)	ZrB ₂ SHS, Si ₃ N ₄ , MAX Phase synthesis	Zr, B, Si precursors
Aerogel Line I	115 (future)	Polyimide-silica aerogel production	SiO ₂ from S-11
Basalt Fiber Furnace	116 (future)	In-house basalt fiber drawing	Basalt rock (direct from AMLS)
BFRP Pultrusion Line	117 (future)	Rebar, profiles, structural shapes	Basalt fiber + Elium®
MXene Reactor	118 (future)	Ti ₃ C ₂ T _x synthesis from MAX Phase	Ti ₃ AlC ₂ (powder 114)
YInMn Kiln	119 (future)	YInMn Blue pigment synthesis	In (S-04), Y (S-10)

5.2 Single Machine Startup (Phase 1 Self-Sufficiency)

In Phase 1, only the Aegis-Forge I is operational. All outputs except direct-use slugs go to: - Commercial sale (Al, Cu, Fe, Zn/Pb/Sn slugs → immediate revenue) - Stockpile for Phase 2 machines (ZrO₂, RE concentrates → await hydromet reactor) - Activated carbon internal use → ad immediately into gas treatment Train C

Phase 1 commercial slug revenue estimate (per 50 kg mixed electronics input):

Material balance for 50 kg input (typical PCB-rich e-waste): - Polymers (35%): 17.5 kg □ S-02 polymer slug □ \$0.15/kg scrap = \$2.63 - Carbon char (8%): 4 kg □ S-03 char □ \$0.10/kg = \$0.40 - Zn/Pb/Sn (6%): 3 kg □ S-05 slug □ \$2.00/kg blended = \$6.00 - Aluminum (12%): 6 kg □ S-06 slug □ \$1.80/kg secondary = \$10.80 - Copper (23%): 11.5 kg □ S-07 billet □ \$9.00/kg = \$103.50 - Precious metals (trace): in S-07 □ Au 15g = \$1,548; Ag: 115g = \$130; Pd: 5g = \$185 = \$1,863 - Iron (14%): 7 kg □ S-09 slug □ \$0.45/kg = \$3.15 - Mineral residue (2%): 1 kg □ S-11 □ retain for CSMFAB - Gas streams □ energy credit □ 18.5 MJ = ~5 kWh electricity equivalent = \$0.85

Revenue per 50 kg batch: \$1,990 (dominated by precious metals) Operating cost per 50 kg batch: electricity (\$45) + N2 (\$2) + consumables (\$15) = \$62 Net per batch: \$1,928

At 8 batches/day (3.5 hr cycle, 12-hr operating day): \$15,424/day gross revenue from e-waste only Annual (250 operating days): \$3.86M/year from Aegis-Forge I alone — exclusively from waste streams!

This revenue directly funds Phase 2 machine construction.

6. Full Machine Specifications — Summary

6.1 Aegis-Forge I Technical Specifications

Parameter	Value
Drum working diameter	600 mm
Drum working length	800 mm
Drum working volume	226 liters
Maximum batch size	80 kg (assumes ~65% fill factor)
Typical batch size	50 kg
Cycle time (full thermal cascade)	3.5-5 hours
Daily throughput (8 batches)	400 kg/day
Annual throughput (250 days)	100 metric tons/year
Maximum drum temperature	2400°C
Maximum drum pressure	5 bar (design); 2.5 bar (operating)
Drum rotational speed	0-500 RPM variable
Maximum centrifugal force	84.5 G at 500 RPM
Induction total installed power	500 kW (8 × 62.5 kW zones)
Acoustic pre-processor power	8 kW LF + 4 kW ultrasonic
Gas treatment N2 loop	Closed-loop, 2-3 kg/cycle makeup
Machine external dimensions	2.2 m (L) × 1.4 m (W) × 1.8 m (H)
Total machine mass	~850 kg
Operating electrical supply	3-phase 480V, 600 A service
Cooling water requirement	15 L/min, 10°C supply
Control system	PMMA POF + FBG + KNbO3-BaTiO3 DSP

6.2 Production Cost Per Machine (Small Batch Production: 5-unit run)

Assembly Group	Material Cost	Labor Cost	Total
ZrB2-SiC Drum (flash sintered)	\$18,500	\$6,200	\$24,700
BFRP Pressure Vessel (filament wound)	\$4,200	\$3,800	\$8,000
Aerogel Insulation Gap Fill	\$1,800	\$400	\$2,200
8-Zone Induction Coil Bank	\$28,000	\$8,500	\$36,500
GaN Induction Inverter Electronics	\$35,000	\$4,200	\$39,200

Assembly Group	Material Cost	Labor Cost	Total
AMLS Acoustic Pre-Processor	\$8,500	\$2,800	\$11,300
Gas Capture System (Trains A-E)	\$22,000	\$9,500	\$31,500
Slug Carousel Mold System	\$6,500	\$3,200	\$9,700
PEEK CF40 Drive Assembly	\$12,000	\$4,800	\$16,800
ZrO2 Bearing and Seal Package	\$5,500	\$2,000	\$7,500
PMMA POF Control System	\$4,800	\$3,500	\$8,300
BFRP Frame and Conveyor	\$3,500	\$2,800	\$6,300
Plasma Torch Unit	\$8,000	\$2,200	\$10,200
Assembly, Testing, Commissioning	\$0	\$18,000	\$18,000
TOTAL	\$158,300	\$71,900	\$230,200

Prototype (1 unit): ~\$385,000 (3× multiplier for NRE + tooling) **Small batch (5 units):** ~\$230,200 each
Production volume (50+ units): ~\$148,000 each **MSRP (production volume):** \$220,000–\$280,000 each
Payback at \$15,424/day revenue: 15-18 days of operation per machine

7. Conclusion: The First Machine Is the Seed

The Aegis-Forge I is not merely a recycling machine. It is an in-situ mining device, a hazardous waste handler, a precious metal recovery system, a raw mineral processor, and a seed material generator simultaneously. By design, it uses exclusively CSMFAB materials — so the first Aegis-Forge I can be assembled using materials already in production.

It produces the materials needed to build the next machine in the assembly line. That machine produces more refined materials needed for the next. And so on — until the full CSMFAB portfolio is self-sustaining from Earth's own raw materials and the circular return of expired Aegis products.

This is the Phoenix Protocol made physical.

End of CSMFAB000000000112 Volume 5 | Assembly Line Integration and Output Slug Catalog End of CSMFAB000000000108-112 Aegis-Forge I Series Carrington Storm Motors | Safe Pod Engineering Company | June 2026